AERO70015/96012 Mathematics 3 Problem Set #2

Problem Sheet 2

2.1 PR2.1

By writing $f(z) = u(x,\ y) + i\ v(x,\ y)$, where $z = x + i\ y$, find the real functions $u(x,\ y)$ and $v(x,\ y)$ in each of the following cases:

(a) $z^2 + 2\ z$

(b) $z/(e^z - 1) \quad (z \neq 0)$

(c) $\sin z$

(d) $1/z \quad (z \neq 0)$

(a) $z^2 + 2z = (x+iy)^2 + 2(x+iy)$

$$= x^2 + 2ixy - y^2 + 2x + 2iy$$

$$= (x^2 - y^2 + 2x) + i(2xy + 2y)$$

(b) $\dfrac{z}{e^z - 1} = \dfrac{x+iy}{e^{x+iy} - 1} = \dfrac{x+iy}{e^x e^{iy} - 1}$

$$= \frac{x+iy}{e^x(\cos y + i\sin y) - 1} = \frac{x+iy}{(e^x\cos y - 1) + ie^x \sin y}$$

$$= \frac{(x+iy)(e^x\cos y - 1 - ie^x\sin y)}{(e^x\cos y - 1)^2 + (e^x\sin y)^2}$$

$$= \frac{xe^x\cos y - x + ye^x\sin y}{e^{2x}(\cos^2 y + \sin^2 y) - 2e^x\cos y + 1} + i\frac{ye^x\cos y - y - xe^x\sin y}{e^{2x}(\cos^2 y + \sin^2 y) - 2e^x\cos y + 1}$$

$$= \frac{xe^x\cos y - x + ye^x\sin y}{e^{2x} - 2e^x\cos y + 1} + i\frac{ye^x\cos y - y - xe^x\sin y}{e^{2x} - 2e^x\cos y + 1}$$

(c) $\sin z = \sin(x+iy) = \sin x\cos iy + \cos x\sin iy = \sin x\cosh y + i\cos x\sinh y$

(d) $\dfrac{1}{z} = \dfrac{1}{x+iy} = \dfrac{x-iy}{(x+iy)(x-iy)}$

$$= \frac{x}{x^2+y^2} + i\left(-\frac{y}{x^2+y^2}\right)$$

2.2 PR2.2

By using the Cauchy-Riemann equations, show that the function

$$f(z) = \left(1 + \frac{i}{4\,\pi}\right)\ \log z$$

is analytic in any region that does not circle the origin.

Show that the equation of any contour on which $u(x,\ y)$ is constant may be written in the form $r = A\ e^{\theta/(4\ \pi)}$, where A is a constant and $(r,\ \theta)$ are polar coordinates. Sketch these contours and also those on which $v(x,\ y)$ is constant.

What physical situation might these contours describe?

$$f(z) = (1 + i\frac{1}{4\pi})\ln(re^{i\theta})$$

$$= (1 + i\frac{1}{4\pi})(\ln r + i\theta)$$

$$= (\ln r - \frac{\theta}{4\pi}) + i(\theta + \frac{1}{4\pi}\ln r)$$

$$\therefore \frac{\partial u}{\partial r} = \frac{1}{r}$$

$$\frac{\partial u}{\partial \theta} = -\frac{1}{4\pi}$$

$$\frac{\partial v}{\partial r} = \frac{1}{4\pi r}$$

$$\frac{\partial v}{\partial \theta} = 1$$

for $r \neq 0$, where $(x, y) \neq (0, 0)$

$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}$ and $\frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}$ $\quad\therefore$ satisfies C-R

$\therefore f(z)$ is analytic despite at origin

Made with Goodnotes

if $u = \ln r - \frac{\theta}{4\pi} = \text{constant} = C$

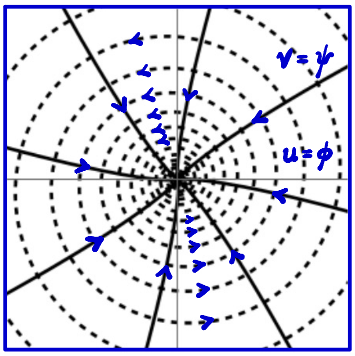
$\ln r = C + \frac{\theta}{4\pi}$

$\therefore r = e^{C + \frac{\theta}{4\pi}} = e^{C} e^{\frac{\theta}{4\pi}} = Ae^{\frac{\theta}{4\pi}}$

similarily, for v, $r = Be^{-4\pi\theta}$

u might be potential function, ϕ

v might be stream function, ψ



flow with sink at origin

2.3 PR2.3 (imported from corrected file)

Using n to stand for any constant integer, and writing π as pi, determine all the possible values of

(a) $\log(1-i)$

(b) $\arctan(2\,i)$

(c) $(1+i)^{2\,i}$

(d) $(1+i)^{1/2}$ (principal value only)

(e) $2^{3+2\,i}$ (principal value only)

(a) $\ln(1-i)$

for $z = re^{i\theta} = 1-i$, $r = \sqrt{2}$, $\theta = \arg z = \tan^{-1}(\frac{-1}{1}) = -\frac{1}{4}\pi + 2n\pi$

$\therefore \ln(1-i) = \ln(re^{i\theta}) = \ln r + i\theta = \ln\sqrt{2} + i(-\frac{1}{4}\pi + 2n\pi)$

(b) $\cancel{\tan^{-1}(2i) = \tan^{-1}[2e^{i(\frac{1}{2}\pi + 2n\pi)}]}$

$z = \tan^{-1}(2i)$

$\therefore \tan z = 2i$

$\therefore \frac{\sin z}{\cos z} = 2i$

$\therefore \frac{e^{iz} - e^{-iz}}{2i} \frac{2}{e^{iz} + e^{-iz}} = \frac{e^{iz} - e^{-iz}}{i(e^{iz} + e^{-iz})} = i \frac{e^{-iz} - e^{iz}}{e^{iz} + e^{-iz}} = 2i$

$\therefore e^{-iz} - e^{iz} = 2e^{iz} + 2e^{-iz}$

$\therefore e^{iz} = -\frac{1}{3} e^{-iz}$

$\therefore e^{2iz} = e^{(2ix - 2y)} = -\frac{1}{3}$

$\therefore e^{-2y}(\cos 2x + i\sin 2x) = -\frac{1}{3}$

$\therefore e^{-2y} = \frac{1}{3}$

$\cos 2x + i\sin 2x = -1$

$\therefore y = -\frac{1}{2}\ln\frac{1}{3} = \ln(\frac{1}{3})^{-\frac{1}{2}} = \ln\sqrt{3}$

$x = \frac{\pi}{2} + n\pi$

$\therefore z = \frac{\pi}{2} + n\pi + i\ln\sqrt{3}$

(c) $(1+i)^{2i} = e^{2i\ln(1+i)}$

according to (a), similarily, $\ln(1+i) = \ln\sqrt{2} + i(\frac{1}{4}\pi + 2n\pi)$

$\therefore (1+i)^{2i} = e^{2i[\ln\sqrt{2} + i(\frac{1}{4}\pi + 2n\pi)]} = e^{\ln 2 i} e^{-\frac{1}{2}\pi - 4n\pi} = e^{-\frac{1}{2}\pi - 4n\pi}(\cos\ln 2 + i\sin\ln 2)$

(d) $(1+i)^{\frac{1}{2}} = e^{\frac{1}{2}\ln(1+i)}$

according to (c), $\ln(1+i) = \ln\sqrt{2} + i(\frac{1}{4}\pi + 2n\pi)$

$\therefore (1+i)^{\frac{1}{2}} = e^{\frac{1}{2}\ln\sqrt{2}} e^{i(\frac{1}{8}\pi + n\pi)} = 2^{\frac{1}{4}} e^{i\frac{1}{8}\pi}$ for principle value $= 2^{\frac{1}{4}}(\cos\frac{1}{8}\pi + i\sin\frac{1}{8}\pi)$

(e) $2^{3+2i} = e^{(3+2i)(\ln 2 + 2n\pi i)}$

$= e^{3\ln 2 - 4n\pi} e^{i(2\ln 2 + 6n\pi)}$

$= 8e^{i\ln 4}$ for principle value

$= 8(\cos\ln 4 + i\sin\ln 4)$

2.4 PR2.4

Evaluate the following using Cauchys integral formula:

(a)

$$\oint_C \frac{e^z}{z + i\,\pi/2}\, dz,$$

where C is the boundary of the square defined by the lines $x = \pm 2$, $y = \pm 2$.

(b)

$$\oint_C \frac{e^z}{z\,(z+1)}\, dz,$$

where C is the boundary of the circle $|z - 1| = 3$.

(a) let $f(z) = e^z$, which is entire

$$\therefore \oint_c \frac{f(z)}{z+i\frac{\pi}{2}} dz = 2\pi i f(-i\tfrac{\pi}{2})$$

the point $z_0 = -i\frac{\pi}{2}$ is inside the path of C

$$\therefore \oint_c \frac{f(z)}{z+i\frac{\pi}{2}} dz = 2\pi i e^{-i\frac{\pi}{2}} = 2\pi i [\cos(\tfrac{\pi}{2}) - i\sin(\tfrac{\pi}{2})] = 2\pi i \cdot (-i) = 2\pi$$

(b) $\oint_c \frac{e^z}{z(z+1)} dz = \oint_c \frac{e^z}{z} - \frac{e^z}{z+1} dz = \oint_c \frac{e^z}{z} dz - \oint_c \frac{e^z}{z+1} dz$

let $f(z) = e^z$, which is entire

the point $(0,0)$ and $(-1,0)$ is inside the path of C

$$\therefore \oint_c \frac{e^z}{z(z+1)} dz = 2\pi i e^0 - 2\pi i e^{-1} = 2\pi(1-e^{-1})i$$

2.5 PR2.5

Evaluate the following integrals, where C is the boundary of the circle $|z| = 2$:

(a)

$$\oint_C \frac{\cos z}{z^n}\, dz, \quad n = 1,\ 2,\ \ldots$$

(b)

$$\oint_C z^n\,(1-z)^m\, dz, \quad m = 0,\ 1,\ 2,\ \ldots; \quad n = 0,\ \pm 1,\ \pm 2,\ \ldots$$

(c)

$$\oint_C \frac{z^3+5}{z-i}\, dz$$

(a) let $f(z) = \cos z$, which is entire

the point $(0,0)$ is inside the path of C

$$\therefore \oint_C \frac{\cos z}{z^n}\, dz = \frac{2\pi i}{(n-1)!}\frac{d^{n-1}}{dz^{n-1}}(\cos z)\Big|_{z=0}$$

$$= \begin{cases} 0, & n = 2, 4, 6, \ldots \\ \frac{2\pi i}{(n-1)!}, & n = 1, 5, 9, \ldots \\ -\frac{2\pi i}{(n-1)!}, & n = 3, 7, 11 \ldots \end{cases}$$

$$= \begin{cases} 0, & n \text{ even} \\ (-1)^{\frac{n-1}{2}}\frac{2\pi i}{(n-1)!}, & n \text{ odd} \end{cases}$$

(b) $\oint z^n (1-z)^m dz$

if $n \geq 0$, $z^n(1-z)^m$ is analytic inside $|z| = 2$

therefore, according to Cauchy's Integral Theorem, $\oint_C z^n(1-z)^m dz = 0$

if $n<0$, let $f(z) = (1-z)^m$, the point $(0,0)$ is inside the path of C

$$\therefore \oint z^n (1-z)^m dz = \oint \frac{(1-z)^m}{z^{-n}} dz = \frac{2\pi i}{(-n-1)!}\frac{d^{-n-1}}{dz^{-n-1}}(1-z)^m\Big|_{z=0}$$

where $\frac{d^r}{dz^r}(1-z)^m = (-1)^r m(m-1)(m-2)\ldots[m-(r-1)](1-z)^{m-r} = (-1)^r\frac{m!}{(m-r)!}(1-z)^{m-r}$ for $m \geq r$

$$\therefore \oint z^n (1-z)^m dz = \begin{cases} 0, & m < -n-1 \\ \frac{2\pi i}{(-n-1)!}(-1)^{-n-1}\frac{m!}{(m+n+1)!}, & m \geq -n-1 \end{cases}$$

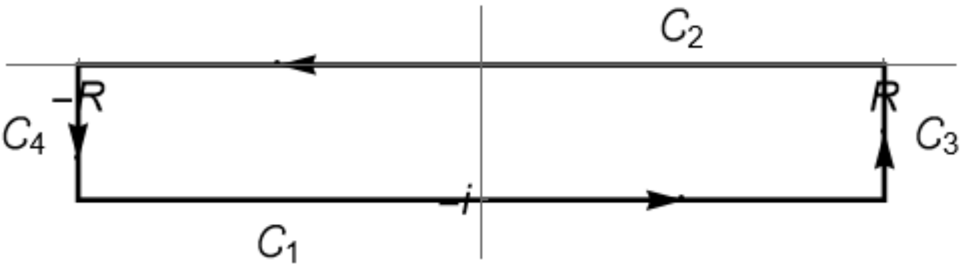
(c) let $f(z) = z^3 + 5$. which is entire.

the point (0, 0) is inside the path of C.

$$\oint_c \frac{z^3+5}{z-i} = 2\pi i\,(i^3+5) = 2\pi i\,(-i+5) = 2\pi + 10\pi i$$

2.6 PR2.6

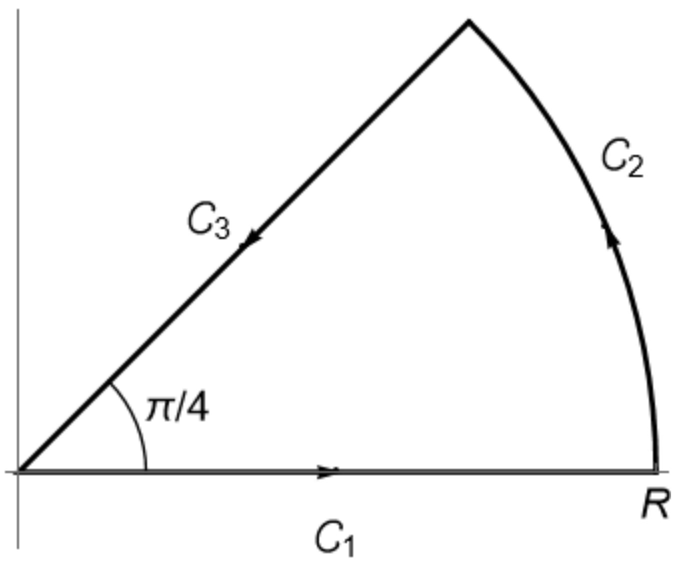
(a)



By considering the integral of $e^{-(z+i)^2}$ along the contour in the figure, and letting R tend to infinity, calculate

$$\int_{-\infty}^{\infty} e^{-x^2-2\,i\,x}\;dx,$$

where you are given that $\int_{-\infty}^{\infty} e^{-x^2}\;dx = \sqrt{\pi}$.



(b)

By considering the integral of $e^{i\,z^2}$ along the contour in the figure, and letting R tend to infinity, calculate

$$\int_{-\infty}^{\infty} e^{i\,x^2}\;dx.$$

You may assume that

$$\int_0^{\pi/4} R\, e^{-R^2 \ \sin(2\ \theta)}\, d\theta$$

tends to zero as R tends to infinity.

(a) $\oint_c e^{-(z+i)^2} dz = \oint_c e^{-z^2-2iz+1} dz = e\oint_c e^{-z^2-2iz} dz$

Also,

$$\oint_c e^{-(z+i)^2} dz = \int_{c_1} e^{-(z+i)^2} dz \Big|_{z=x-i} + \int_{c_2} e^{-(z+i)^2} dz \Big|_{z=x} + \int_{c_3} e^{-(z+i)^2} dz \Big|_{z=R+y} + \int_{c_4} e^{-(z+i)^2} dz \Big|_{z=-R+y}$$

$$= \int_{-\infty}^{\infty} e^{-x^2} dx + \int_{\infty}^{-\infty} e^{-(x+i)^2} dx + \int_{-i}^{0} e^{-\infty} dy + \int_{0}^{-i} e^{-\infty} dy$$

$$= \sqrt{\pi} - e\int_{-\infty}^{\infty} e^{-x^2-2ix} dx + 0 - 0$$

According to Cauchy's Integral Theorem, since $e^{-(z+i)}$ is entire, $\oint_c e^{-(z+i)^2} dz = 0$

$$\therefore \int_{-\infty}^{\infty} e^{-x^2-2ix} dx = \frac{\sqrt{\pi}}{e}$$

(b) According to Cauchey's Integral Theorem, since e^{iz^2} is entire

$$\oint_c e^{iz^2} dz = 0$$

Also,

$$\oint_c e^{iz^2} dz = \int_{c_1} e^{iz^2} dz \Big|_{z=x} + \int_{c_2} e^{iz^2} dz \Big|_{z=Re^{i\theta}} + \int_{c_3} e^{iz^2} dz \Big|_{z=t e^{i\frac{\pi}{4}}}$$

$$= \int_0^R e^{ix^2} dx + \int_0^{\frac{\pi}{4}} e^{iR^2(\cos 2\theta + i\sin 2\theta)}\, iRe^{i\theta} d\theta + \int_{R}^{0} \overset{e^{-t^2}e^{i\frac{\pi}{4}}}{e^{it}}\, dt$$

$$= \int_0^R e^{ix^2} dx + \int_0^{\frac{\pi}{4}} iRe^{iR^2\cos 2\theta - R^2\sin 2\theta}(\cos\theta + i\sin\theta)\, d\theta - \int_0^R e^{-t^2} e^{i\frac{\pi}{4}} dt$$

where $\left| \int_0^{\frac{\pi}{4}} iRe^{iR^2\cos 2\theta - R^2\sin 2\theta}(\cos\theta + i\sin\theta)\, d\theta \right| \leq |i| \int_0^{\frac{\pi}{4}} |R| \underbrace{|e^{iR^2\cos 2\theta}|}_{=1} |e^{-R^2\sin 2\theta}| \underbrace{|(\cos\theta + i\sin\theta)|}_{=1} d\theta$

$$= i\int_0^{\frac{\pi}{4}} Re^{-R^2\sin 2\theta} d\theta \longrightarrow 0 \quad \text{as } R \longrightarrow \infty$$

$$\therefore \int_0^{\frac{\pi}{4}} iRe^{iR^2\cos 2\theta - R^2\sin 2\theta}(\cos\theta + i\sin\theta)\, d\theta \longrightarrow 0 \quad \text{as } R \longrightarrow \infty$$

and $\int_0^R e^{-t^2} e^{i\frac{\pi}{4}} dt = e^{i\frac{\pi}{4}} \int_0^R e^{-t^2} dt = e^{i\frac{\pi}{4}} (\frac{1}{2}\sqrt{\pi})$ as $R \to \infty$

$\therefore \int_0^R e^{ix^2} dx = \sqrt{\pi} e^{i\frac{\pi}{4}}$

2.7 Past Exam Question Part, 2017

(a) The complex function $f(z)$ is given by

$$f(z) = u(x, y) + i\, v(x, y),$$

where x and y are, respectively, the real and imaginary parts of z, and u and v are differentiable functions from $\mathbb{R}^2$ to $\mathbb{R}$.

By considering the two limits as $\delta z \to 0$ of

$$\frac{f(z_0 + \delta z) - f(z_0)}{\delta z}$$

as the point $z_0 = x_0 + i\, y_0$ is approached along paths parallel to the real axis and the imaginary axis, prove that if $f(z)$ is complex-differentiable at z_0, then at $(x_0,\ y_0)$ u and v satisfy the Cauchy-Riemann equations,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y},$$
$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}.$$

(b) Find, in terms of the real variables x and y, the real and imaginary parts, u and v respectively, of $(x + i\, y)^3$, and verify that they satisfy the Cauchy-Riemann equations everywhere.

By considering the partial derivatives $\partial u/\partial x$ and $\partial v/\partial x$, show that the complex derivative of z^3 is $3\, z^2$.

(a) when along the path parallel to the real axis.

$$\frac{\partial f}{\partial z} = \frac{u(x_0+\delta x, y_0) - u(x_0, y_0) + i[v(x_0+\delta x, y_0) - v(x_0, y_0)]}{\delta x} = \frac{\delta u}{\delta x} + i\frac{\delta v}{\delta x}$$

when along the path parallel to the imaginary axis.

$$\frac{\partial f}{\partial z} = \frac{u(x_0, y_0+\delta y) - u(x_0, y_0) + i[v(x_0, y_0+\delta y) - v(x_0, y_0)]}{i\delta y} = \frac{\delta v}{\delta y} - i\frac{\delta u}{\delta y}$$

equate both equation. therefore. get

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

(b) $(x+iy)^3 = (x^2 + 2ixy - y^2)(x+iy) = x^3 + 2ix^2y - xy^2 + ix^2y - 2xy^2 - iy^3$

$$= (x^3 - 3xy^2) + i(2x^2y + x^2y - y^3)$$

$\therefore u = x^3 - 3xy^2$

$v = 2x^2y + x^2y - y^3$

$\therefore \frac{\partial u}{\partial x} = 3x^2 - 3y^2$

$\frac{\partial u}{\partial y} = -6xy$

$\frac{\partial v}{\partial x} = 4xy + 2xy = 6xy$

$\frac{\partial v}{\partial y} = 2x^2 + x^2 - 3y^2 = 3x^2 - 3y^2$

$\therefore \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

$\therefore$ satisfied C-R equation, $(x+iy)^3$ is entire

$\therefore \frac{\partial f}{\partial z} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}$

$= 3x^2 - 3y^2 + 6ixy$

$= 3(x+iy)^2$